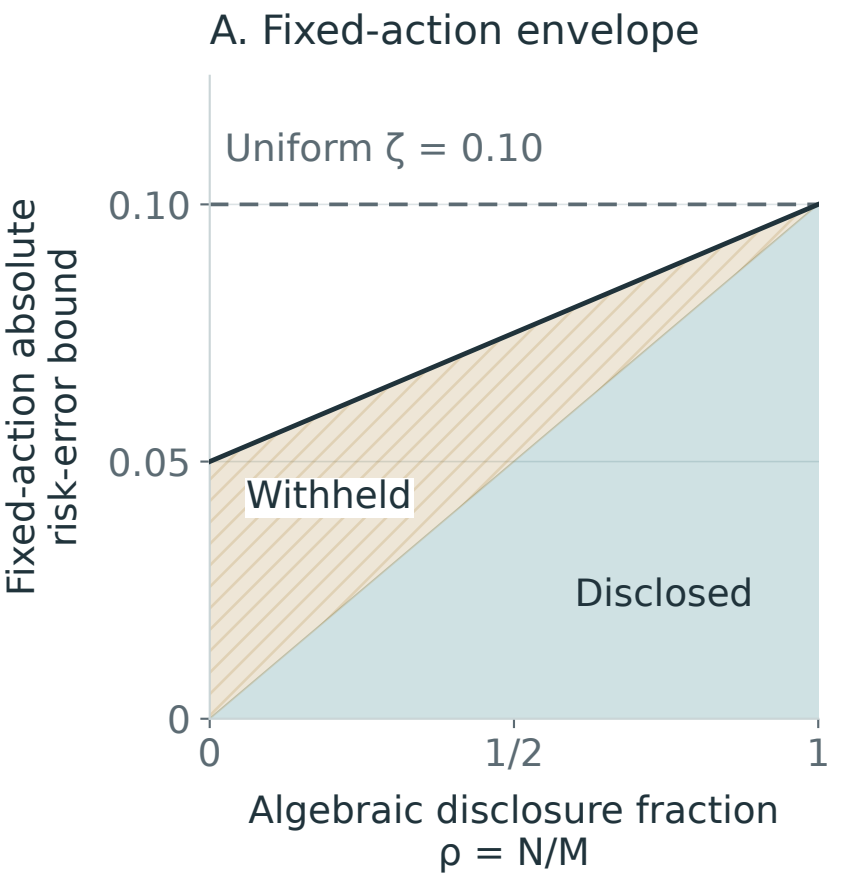
# Two coefficient uncertainties contribute to risk

τ = 0.25; Δb = 0.10; Δd = 0.20.
Illustrative simultaneous radii.




Disclosed: $\rho\Delta_b$
Withheld: $(1-\rho)\tau\Delta_d$
Total $B(\rho) = 0.05 + 0.05\rho$

B(0) = 0.05: withheld distinctness.
B(1) = 0.10: disclosed direction.

## B. A separate regret argument

Same nonempty finite family F.
g*: exact minimizer.
ĝ: estimated optimizer (η = 0).

**Exact → estimated at ĝ**

$$R(\hat{g}) - \widehat{R}(\hat{g}) \le \zeta = 0.10$$

+

**Estimated optimization**

$$\widehat{R}(\hat{g}) - \widehat{R}(g^*) \le \eta = 0$$

+

**Estimated → exact at g***

$$\widehat{R}(g^*) - R(g^*) \le \zeta = 0.10$$

**Exact-risk regret** $\le \zeta + \eta + \zeta = 0.20$

Algebraic bound; no attained action path.
Feasible ρ depends on n and the family.
Risk and regret use the same M and loss.